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MOLDED FLOW CHANNEL

Abstract

A nanopore based sequencing system includes a plurality of nanopore sensors. Each nanopore sensor has a portion for receiving a fluid. The nanopore based sequencing system includes a fluid chamber configured to guide the fluid over the plurality of nanopore sensors and an inlet configured to deliver the fluid into the fluid chamber. At least a portion of the fluid chamber is made of a material that has been molded around at least a portion of an electrode.

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Background/Summary

CROSS REFERENCE TO OTHER APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/316,920, filed May 12, 2023, which is a continuation of U.S. patent application Ser. No. 16/947,043, filed Jul. 15, 2020, now U.S. Pat. No. 11,686,706, which is a continuation of U.S. patent application Ser. No. 15/410,586, filed Jan. 19, 2017, now U.S. Pat. No. 10,718,737, entitled MOLDED FLOW CHANNEL, which claims priority to U.S. Provisional Patent Application No. 62/281,663 entitled CHIP PERIMETER SEAL AND BOND filed Jan. 21, 2016, and claims priority to U.S. Provisional Patent Application No. 62/281,662 entitled USE OF TITANIUM NITRIDE AS AN INTERMEDIATE LAYER IN BONDING GLASS TO A SILICON CHIP filed Jan. 21, 2016, and claims priority to U.S. Provisional Patent Application No. 62/286,826 entitled USE OF TITANIUM NITRIDE AS AN INTERMEDIATE LAYER IN BONDING GLASS TO A SILICON CHIP filed Jan. 25, 2016, all of which are incorporated herein by reference for all purposes.

BACKGROUND OF THE INVENTION

[0002] Advances in micro-miniaturization within the semiconductor industry in recent years have enabled biotechnologists to begin packing traditionally bulky sensing tools into smaller and smaller form factors, onto so-called biochips. It would be desirable to develop techniques for biochips that make them more robust, efficient, and cost-effective.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Various embodiments of the invention are disclosed in the following detailed description and the accompanying drawings.

[0004] FIG. 1 illustrates an embodiment of a cell **100** in a nanopore based sequencing chip.

[0005] FIG. 2 illustrates an embodiment of a cell **200** performing nucleotide sequencing with the Nano-SBS technique.

[0006] FIG. 3 illustrates an embodiment of a cell about to perform nucleotide sequencing with pre-loaded tags.

[0007] FIG. 4 illustrates an embodiment of a process **400** for nucleic acid sequencing with pre-loaded tags.

[0008] FIG. 5 illustrates an embodiment of a fluidic workflow process **500** for flowing different types of liquids or gases through the cells of a nanopore based sequencing chip during different phases of the chip's operation.

[0009] FIG. 6A illustrates an exemplary flow of a liquid or gas across the nanopore based sequencing chip.

[0010] FIG. 6B illustrates another exemplary flow of a liquid or gas across the nanopore based sequencing chip.

[0011] FIG. 7A illustrates an exemplary flow of a first type of fluid across the nanopore based sequencing chip.

[0012] FIG. 7B illustrates that a second fluid is flowed through the chip after a first fluid has been flowed through the chip at an earlier time.

[0013] FIG. 8 illustrates the top view of a nanopore based sequencing system **800** with a flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface.

[0014] FIG. 9 illustrates the various components that are assembled together to form the nanopore based sequencing system **800** as shown in FIG. 8.

[0015] FIG. 10 illustrates another exemplary view of nanopore based sequencing system **800**.

[0016] FIG. 11A illustrates the top view of a nanopore based sequencing system **1100** with an improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface.

[0017] FIG. 11B illustrates the cross sectional view of system **1100** from the position of a plane **1114** through the system.

[0018] FIG. 12A illustrates another exemplary view of nanopore based sequencing system **1100** with a fan-out plenum.

[0019] FIG. 12B illustrates the various components that are assembled together to form nanopore based sequencing system **1100** as shown in FIG. 11.

[0020] FIG. 13 illustrates the paths that are followed by a fluid as it flows through the nanopore based sequencing system **1100** with a fan-out plenum.

[0021] FIG. 14 illustrates the top view of a nanopore based sequencing system **1400** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface.

[0022] FIG. 15 illustrates the top view of a nanopore based sequencing system **1500** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface.

[0023] FIG. 16 illustrates the top view of a nanopore based sequencing system **1600** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface.

[0024] FIG. 17 illustrates the top view of a nanopore based sequencing system **1700** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface.

[0025] FIG. 18 illustrates the top view of a nanopore based sequencing system **1800** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface.

[0026] FIG. 19A illustrates an exemplary view of one embodiment of a nanopore based sequencing system **1900** with a serpentine flow channel.

[0027] FIG. 19B illustrates the various components that are laminated together to form nanopore based sequencing system **1900**.

[0028] FIG. 20A illustrates the top side view of a backing plate and a flexible flat circuit that is connected to the counter electrode (not visible) located on the bottom side of the backing plate.

[0029] FIG. 20B illustrates the same unit **2000** as shown in FIG. 20A when the backing plate is flipped upside down.

[0030] FIG. 20C illustrates the various components of unit **2000** that are laminated together.

[0031] FIG. 21A illustrates a cross sectional view of a flow channel **2100** with sharp edges or sharp corners that may trap fluids more easily.

[0032] FIG. 21B illustrates a cross sectional view of a flow channel **2102** that has a D-shaped cross sectional geometry.

[0033] FIG. 21C illustrates a cross sectional view of another flow channel **2106** that has a D-

shaped cross sectional geometry.

[0034] FIG. **22** illustrates a side view of a nanopore based sequencing system with flow channels having a D-shaped cross sectional geometry.

[0035] FIG. **23** is a diagram illustrating an embodiment of a molded flow channel component.

[0036] FIG. **24** is a diagram illustrating an embodiment of a counter electrode insert.

[0037] FIG. **25** is a diagram illustrating an embodiment of a mold for a molded flow channel component.

[0038] FIG. **26** is a diagram illustrating an embodiment of a molded flow channel component removed from a mold.

[0039] FIG. **27** is a diagram illustrating a portion of an embodiment of a nanopore based sequencing system utilizing a molded flow channel component.

[0040] FIG. **28** is a diagram illustrating an embodiment of a clamping of nanopore based sequencing system components.

[0041] FIG. **29** is a diagram illustrating an embodiment of encapsulated wire bonds.

[0042] FIG. **30** is a diagram illustrating an embodiment of encapsulating wire bonds together with one or more flow channel components in a single manufacturing step.

DETAILED DESCRIPTION

[0043] The invention can be implemented in numerous ways, including as a process; an apparatus; a system; a composition of matter; a computer program product embodied on a computer readable storage medium; and/or a processor, such as a processor configured to execute instructions stored on and/or provided by a memory coupled to the processor. In this specification, these implementations, or any other form that the invention may take, may be referred to as techniques. In general, the order of the steps of disclosed processes may be altered within the scope of the invention. Unless stated otherwise, a component such as a processor or a memory described as being configured to perform a task may be implemented as a general component that is temporarily configured to perform the task at a given time or a specific component that is manufactured to perform the task. As used herein, the term ‘processor’ refers to one or more devices, circuits, and/or processing cores configured to process data, such as computer program instructions.

[0044] A detailed description of one or more embodiments of the invention is provided below along with accompanying figures that illustrate the principles of the invention. The invention is described in connection with such embodiments, but the invention is not limited to any embodiment. The scope of the invention is limited only by the claims and the invention encompasses numerous alternatives, modifications and equivalents. Numerous specific details are set forth in the following description in order to provide a thorough understanding of the invention. These details are provided for the purpose of example and the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured.

[0045] Nanopore membrane devices having pore sizes on the order of one nanometer in internal diameter have shown promise in rapid nucleotide sequencing. When a voltage potential is applied across a nanopore immersed in a conducting fluid, a small ion current attributed to the conduction of ions across the nanopore can be observed. The size of the current is sensitive to the pore size.

[0046] A nanopore based sequencing chip may be used for DNA sequencing. A nanopore based sequencing chip incorporates a large number of sensor cells configured as an array. For example, an array of one million cells may include 1000 rows by 1000 columns of cells.

[0047] FIG. **1** illustrates an embodiment of a cell **100** in a nanopore based sequencing chip. A membrane **102** is formed over the surface of the cell. In some embodiments, membrane **102** is a lipid bilayer. The bulk electrolyte **114** containing protein nanopore transmembrane molecular complexes (PNTMC) and the analyte of interest is placed directly onto the surface of the cell. A single PNTMC **104** is inserted into membrane **102** by electroporation. The individual membranes

in the array are neither chemically nor electrically connected to each other. Thus, each cell in the array is an independent sequencing machine, producing data unique to the single polymer molecule associated with the PNTMC. PNTMC **104** operates on the analytes and modulates the ionic current through the otherwise impermeable bilayer.

[0048] With continued reference to FIG. **1**, analog measurement circuitry **112** is connected to a metal electrode **110** covered by a thin film of electrolyte **108**. The thin film of electrolyte **108** is isolated from the bulk electrolyte **114** by the ion-impermeable membrane **102**. PNTMC **104** crosses membrane **102** and provides the only path for ionic current to flow from the bulk liquid to working electrode **110**. The cell also includes a counter electrode (CE) **116**, which is an electrochemical potential sensor. The cell also includes a reference electrode **117**.

[0049] In some embodiments, a nanopore array enables parallel sequencing using the single molecule nanopore based sequencing by synthesis (Nano-SBS) technique. FIG. **2** illustrates an embodiment of a cell **200** performing nucleotide sequencing with the Nano-SBS technique. In the Nano-SBS technique, a template **202** to be sequenced and a primer are introduced to cell **200**. To this template-primer complex, four differently tagged nucleotides **208** are added to the bulk aqueous phase. As the correctly tagged nucleotide is complexed with the polymerase **204**, the tail of the tag is positioned in the barrel of nanopore **206**. The tag held in the barrel of nanopore **206** generates a unique ionic blockade signal **210**, thereby electronically identifying the added base due to the tags' distinct chemical structures.

[0050] FIG. **3** illustrates an embodiment of a cell about to perform nucleotide sequencing with pre-loaded tags. A nanopore **301** is formed in a membrane **302**. An enzyme **303** (e.g., a polymerase, such as a DNA polymerase) is associated with the nanopore. In some cases, polymerase **303** is covalently attached to nanopore **301**. Polymerase **303** is associated with a nucleic acid molecule **304** to be sequenced. In some embodiments, the nucleic acid molecule **304** is circular. In some cases, nucleic acid molecule **304** is linear. In some embodiments, a nucleic acid primer **305** is hybridized to a portion of nucleic acid molecule **304**. Polymerase **303** catalyzes the incorporation of nucleotides **306** onto primer **305** using single stranded nucleic acid molecule **304** as a template. Nucleotides **306** comprise tag species ("tags") **307**.

[0051] FIG. **4** illustrates an embodiment of a process **400** for nucleic acid sequencing with pre-loaded tags. At stage A, a tagged nucleotide (one of four different types: A, T, G, or C) is not associated with the polymerase. At stage B, a tagged nucleotide is associated with the polymerase. At stage C, the polymerase is in close proximity to the nanopore. The tag is pulled into the nanopore by an electrical field generated by a voltage applied across the membrane and/or the nanopore.

[0052] Some of the associated tagged nucleotides are not base paired with the nucleic acid molecule. These non-paired nucleotides typically are rejected by the polymerase within a time scale that is shorter than the time scale for which correctly paired nucleotides remain associated with the polymerase. Since the non-paired nucleotides are only transiently associated with the polymerase, process **400** as shown in FIG. **4** typically does not proceed beyond stage B.

[0053] Before the polymerase is docked to the nanopore, the conductance of the nanopore is ~300 pico Siemens (300 pS). At stage C, the conductance of the nanopore is about 60 pS, 80 pS, 100 pS, or 120 pS corresponding to one of the four types of tagged nucleotides. The polymerase undergoes an isomerization and a transphosphorylation reaction to incorporate the nucleotide into the growing nucleic acid molecule and release the tag molecule. In particular, as the tag is held in the nanopore, a unique conductance signal (e.g., see signal **210** in FIG. **2**) is generated due to the tag's distinct chemical structures, thereby identifying the added base electronically. Repeating the cycle (i.e., stage A through E or stage A through F) allows for the sequencing of the nucleic acid molecule. At stage D, the released tag passes through the nanopore.

[0054] In some cases, tagged nucleotides that are not incorporated into the growing nucleic acid molecule will also pass through the nanopore, as seen in stage F of FIG. **4**. The unincorporated

nucleotide can be detected by the nanopore in some instances, but the method provides a means for distinguishing between an incorporated nucleotide and an unincorporated nucleotide based at least in part on the time for which the nucleotide is detected in the nanopore. Tags bound to unincorporated nucleotides pass through the nanopore quickly and are detected for a short period of time (e.g., less than 10 ms), while tags bound to incorporated nucleotides are loaded into the nanopore and detected for a long period of time (e.g., at least 10 ms).

[0055] FIG. 5 illustrates an embodiment of a fluidic workflow process **500** for flowing different types of fluids (liquids or gases) through the cells of a nanopore based sequencing chip during different phases of the chip's operation. The nanopore based sequencing chip operates in different phases, including an initialization and calibration phase (phase **502**), a membrane formation phase (phase **504**), a nanopore formation phase (phase **506**), a sequencing phase (phase **508**), and a cleaning and reset phase (phase **510**).

[0056] At the initialization and calibration phase **502**, a salt buffer is flowed through the cells of the nanopore based sequencing chip at **512**. The salt buffer may be potassium chloride (KCl), potassium acetate (KAc), sodium trifluoroacetate (NaTFA), and the like.

[0057] At the membrane formation phase **504**, a membrane, such as a lipid bilayer, is formed over each of the cells. At **514**, a lipid and decane mixture is flowed over the cells. At **516**, a salt buffer is flowed over the cells. At **518**, voltage measurements across the lipid bilayers are made to determine whether the lipid bilayers are properly formed. If it is determined that the lipid bilayers are not properly formed, then step **516** is repeated; otherwise, the process proceeds to step **520**. At **520**, a salt buffer is again introduced.

[0058] At the nanopore formation phase **506**, a nanopore is formed in the bilayer over each of the cells. At **522**, a sample and a pore/polymerase mixture are flowed over the cells.

[0059] At the sequencing phase **508**, DNA sequencing is performed. At **524**, StartMix is flowed over the cells, and the sequencing information is collected and stored. StartMix is a reagent that initiates the sequencing process. After the sequencing phase, one cycle of the process is completed at **526**.

[0060] At the cleaning and reset phase **510**, the nanopore based sequencing chip is cleaned and reset such that the chip can be recycled for additional uses. At **528**, a surfactant is flowed over the cells. At **530**, ethanol is flowed over the cells. In this example, a surfactant and ethanol are used for cleaning the chip. However, alternative fluids may be used. Steps **528** and **530** may also be repeated a plurality of times to ensure that the chip is properly cleaned. After step **530**, the lipid bilayers and pores have been removed and the fluidic workflow process **500** can be repeated at the initialization and calibration phase **502** again.

[0061] As shown in process **500** described above, multiple fluids with significantly different properties (e.g., compressibility, hydrophobicity, and viscosity) are flowed over an array of sensors on the surface of the nanopore based sequencing chip. For improved efficiency, each of the sensors in the array should be exposed to the fluids or gases in a consistent manner. For example, each of the different types of fluids should be flowed over the nanopore based sequencing chip such that the fluid or gas may be delivered to the chip, evenly coating and contacting each of the cells' surface, and then delivered out of the chip. As described above, a nanopore based sequencing chip incorporates a large number of sensor cells configured as an array. As the nanopore based sequencing chip is scaled to include more and more cells, achieving an even flow of the different types of fluids or gases across the cells of the chip becomes more challenging.

[0062] FIG. 6A illustrates an exemplary flow of a fluid across the nanopore based sequencing chip. In FIG. 6A, an inlet (e.g., a tube) **604** delivers a fluid to a nanopore based sequencing chip **602**, and an outlet **606** delivers the fluid or gas out of the chip. Due to the difference in width between the inlet and the nanopore based sequencing chip, as the fluid or gas enters chip **602**, the fluid or gas flows through paths that cover the cells that are close to the outer perimeter but not the cells in the center portion of the chip.

[0063] FIG. 6B illustrates another exemplary flow of a fluid across the nanopore based sequencing chip. In FIG. 6B, an inlet **610** delivers a fluid to a nanopore based sequencing chip **608**, and an outlet **612** delivers the fluid or gas out of the chip. As the fluid or gas enters chip **608**, the fluid or gas flows through paths that cover the cells that are close to the center portion of the chip but not the cells that are close to the outer perimeter of the chip.

[0064] As shown in FIG. 6A and FIG. 6B above, the nanopore based sequencing chip has one or more “dead” zones in the flow chamber. In the embodiment shown in FIG. 6A, the dead zones are distributed close to the center of the chip. In the embodiment shown in FIG. 6B, the dead zones are distributed close to the outer perimeter of the chip. The sensors in the chip array beneath the dead zones are exposed to a small amount of the fluid or a slow flow of the fluid, while the sensors outside of the dead zones are exposed to an excess or fast flow of the fluid.

[0065] Furthermore, the introduction of a second fluid may not displace the first fluid in the dead zones effectively. FIG. 7A illustrates an exemplary flow of a first type of fluid across the nanopore based sequencing chip. In this example, since the dead zones are located at the corners of the nanopore base sequencing chip, the corners of the chip are exposed to the first fluid later than other portions of the chip, but eventually the corners are finally filled up with the first fluid. FIG. 7B illustrates that a second fluid is flowed through the chip after a first fluid has been flowed through the chip at an earlier time. Because the dead zones are located at the corners of the chip, the second fluid fails to displace the first fluid at the corners within a short period of time. As a result, the sensors in the array are not exposed to the right amount of fluid in a consistent manner.

[0066] The design of the flow chamber may also affect the formation of lipid bilayers with the appropriate thickness. With reference to step **514** of process **500** in FIG. 5, a lipid and decane mixture is flowed over the cells, creating a thick lipid layer on top of each of the cells. In order to reduce the thickness of a lipid layer, in some embodiments, one or more air bubbles are flowed over the sensor to scrape the lipid layer into a thinner layer at step **516** of process **500**. The design of the flow chamber should be optimized to control the scraping boundary between the air and the lipid layers, such that an even wiping action is performed over all of the sensors. In addition, the design of the flow chamber may be optimized to prevent the air bubbles from collapsing mid-way across the flow chamber; otherwise, only a portion of the lipid layers in the chip are scraped or “thinned.”

[0067] With continued reference to FIG. 6 and FIG. 7, when the flow chamber flows the fluid from one end to the opposite end of the chip, the size of the dead zones within the chip and the collapsing of the air bubbles, in some embodiments, may be reduced by controlling the flow of the fluids and the air bubbles using different pressure and velocity. However, the improvement is limited.

[0068] FIG. 8 illustrates the top view of a nanopore based sequencing system **800** with a flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface. In this example, the nanopore array chip **802** includes 16 sensor banks (**804**) in a 4×4 row-column arrangement. However, other arrangements of the sensors cells may be used as well. System **800** includes a counter electrode **812** positioned above the flow chamber. Fluids are directed from an inlet **806** to the flow chamber atop chip **802**, and the fluids are directed out of the flow chamber via an outlet **808**. The inlet and the outlet may be tubes or needles. Inlet **806** and outlet **808** are each positioned at one of two corners of the nanopore array chip **802** diagonally across from each other. Because the chamber is considerably wider than the inlet's width, as the fluid or gas enters the chamber, the fluid or gas flows through different paths **810** that cover more cells that are close to the center portion of the chip than cells that are close to the remaining two corners of the chip. The fluid or gas travels from one corner to another diagonal corner, leaving trapped fluids in dead zones in the remaining corners.

[0069] FIG. 9 illustrates the various components that are assembled together to form the nanopore based sequencing system **800** as shown in FIG. 8. System **800** includes various components,

including a printed circuit board **902**, a nanopore array chip **802**, a gasket **904**, counter and reference electrodes **906** connected by a flexible flat circuit **910** to a connector **912**, a top cover **914**, an inlet/outlet guide **916**, an inlet **806**, and an outlet **808**.

[0070] FIG. **10** illustrates another exemplary view of nanopore based sequencing system **800**. The flow chamber is the space formed between the top cover **914**, the gasket **904**, and the nanopore array chip **802**. The chamber volume is shown as **1002** in FIG. **10**.

[0071] FIG. **11A** illustrates the top view of a nanopore based sequencing system **1100** with an improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface. FIG. **11B** illustrates the cross sectional view of system **1100** from the position of a plane **1114** through the system.

[0072] A fluid is directed into system **1100** through an inlet **1102**. Inlet **1102** may be a tube or a needle. For example, the tube or needle may have a diameter of one millimeter. Instead of feeding the fluid or gas directly into the flow chamber, inlet **1102** feeds the fluid or gas to a fan-out plenum space or reservoir **1106**. As shown in the top view of system **1100** (FIG. **11A**), fan-out plenum **1106** directs the fluid or gas outwardly from a central point, a small orifice **1118** of inlet **1102** that intersects (see FIG. **11B**) with the fan-out plenum **1106**. Fan-out plenum **1106** spreads out from orifice **1118** into a fanlike shape. For example, the fanlike shape as shown in FIG. **11A** is a substantially triangular shape. However, other similar shapes that direct the fluid or gas outwardly from the small orifice **1118** may be used as well. In one example, orifice **1118** is one millimeter wide, and fan-out plenum **1106** fans out to seven millimeters, the width of one row of four sensor banks **1122**.

[0073] With reference to the cross sectional view of system **1100** (FIG. **11B**), the fluid or gas fills fan-out plenum **1106** first and then spills over and drains down a narrow slit or slot **1108** that intersects with a flow chamber **1116**, like a waterfall. Flow chamber **1116** allows the fluid or gas to pass over and contact sensors on the surface of nanopore array chip **1120**. Because slit **1108** spans across a row of sensor banks **1122**, the fluid or gas is flowed more evenly across the sensor cells, reducing the number and areas of the dead zones within the chip. As the fluid or gas sweeps across the chip, the fluid or gas reaches a second narrow slit **1112** at the opposite end of the chip, and the fluid or gas is directed through slit **1112** up to a reverse fan-out plenum **1110**. Reverse fan-out plenum **1110** directs the fluid or gas towards a central point, a small orifice **1119** of outlet **1104** that intersects (see FIG. **11B**) with the reverse fan-out plenum **1110**. The fluid or gas is then directed out of system **1100** via an outlet **1104**.

[0074] FIG. **12A** illustrates another exemplary view of nanopore based sequencing system **1100** with a fan-out plenum. FIG. **12B** illustrates the various components that are assembled together to form nanopore based sequencing system **1100** as shown in FIG. **11**. System **1100** includes various components, including a printed circuit board **1201**, a nanopore array chip **1120**, a gasket **1202**, a gasket cover **1204**, a middle plate **1206**, a middle plate **1208**, a reference electrode **1214**, a middle plate **1210**, a counter electrode **1218**, a reference electrode **1216**, a top plate **1212**, an inlet **1102**, and an outlet **1104**.

[0075] The fan-out plenum is the space formed between top plate **1212**, a fan-out void **1220** on the middle layer **1210**, and middle layer **1208**. Slit **1108** is the space formed by aligning a slit **1108A** on middle plate **1208**, a slit **1108B** on middle plate **1206**, and a slit **1108C** on gasket cover **1204**, and stacking middle plate **1208**, middle plate **1206**, and gasket cover **1204** on top of each other. The flow chamber is the space formed between gasket cover **1204**, gasket **1202**, and the nanopore array chip **1120**.

[0076] FIG. **13** illustrates the paths that are followed by a fluid as it flows through the nanopore based sequencing system **1100** with a fan-out plenum. A fluid flows down inlet **1102** (see path **1302A**), fills fan-out plenum **1106** first (see path **1302B**) and then spills over and drains down slit **1108** (see path **1302C**) that intersects with the flow chamber. The flow chamber allows the fluid or gas to pass over and contact sensors on the surface of the nanopore array chip as shown in path

1302D. Because slit **1108** spans across a row of sensor banks, the fluid or gas is flowed more evenly across the sensor cells, reducing the number and areas of the dead zones within the chip. As the fluid or gas sweeps across the chip, the fluid or gas reaches slit **1112** at the opposite end of the chip, and the fluid or gas is directed up through slit **1112** (see path **1302E**) to a reverse fan-out plenum **1110**. Reverse fan-out plenum **1110** converges the fluid or gas towards a central point (see path **1302F**), a small orifice of outlet **1104** that intersects with reverse fan-out plenum **1110**. The fluid or gas is then directed out of system **1100** via an outlet **1104** as shown in path **1302G**.

[0077] FIG. **14** illustrates the top view of a nanopore based sequencing system **1400** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface. The flow chamber is divided into multiple channels **1408**, each channel **1408** directing the fluids to flow directly above a single column (or a single row) of sensor banks **1406**. As shown in FIG. **14**, system **1400** includes four inlets **1402** and four outlets **1404**.

[0078] With reference to FIG. **14**, a fluid is directed into system **1400** in parallel through the four inlets **1402**. Inlet **1402** may be a tube or a needle. For example, the tube or needle may have a diameter of one millimeter. Instead of feeding the fluid or gas directly into a wide flow chamber with a single continuous space, each of the inlets **1402** feeds the fluid or gas into a separate channel **1408** that directs the fluid or gas to flow directly above a single column of sensor banks **1406**. The channels **1408** may be formed by stacking together a top plate and a gasket with dividers **1410** that divide the chamber into channels, and then mounting them on top of the chip. Once the fluid or gas flows through the channels **1408** to the opposite side of the chip, the fluid or gas is directed up in parallel through the four outlets **1404** and out of system **1400**.

[0079] FIG. **15** illustrates the top view of a nanopore based sequencing system **1500** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface. Similar to system **1400**, the flow chamber in system **1500** is divided into multiple channels **1502**, but each channel **1502** directs the fluids to flow directly above two columns (or two rows) of sensor banks **1504**. The width of the channels is about 3.5 millimeters. As shown in FIG. **15**, system **1500** includes two inlets **1506** and two outlets **1508**.

[0080] Both system **1400** and system **1500** allow the fluids to flow more evenly on top of all the sensors on the chip surface. The channel width is configured to be narrow enough such that capillary action has an effect. More particularly, the surface tension (which is caused by cohesion within the fluid) and adhesive forces between the fluid and the enclosing surfaces act to hold the fluid together, thereby preventing the fluid or the air bubbles from breaking up and creating dead zones. Therefore, when the width of a sensor bank is narrow enough, each of the flow channels may flow the fluids directly above two or more columns (or two or more rows) of sensor banks. In this case, system **1500** may be used. When the width of a sensor bank is not narrow enough, then each of the flow channels may flow the fluids directly above one column (or one row) of sensor banks only. In this case, system **1400** may be used.

[0081] FIG. **16** illustrates the top view of a nanopore based sequencing system **1600** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface. The flow chamber is divided into two horseshoe-shaped flow channels **1608**, each channel **1608** directing the fluids to flow directly above a single column (or a single row) of sensor banks **1606** from one end of the chip to the opposite end and then directing the fluids to loop back and flow directly above a second adjacent column of sensor banks to the original end of the chip. As shown in FIG. **16**, system **1600** includes two inlets **1602** and two outlets **1604**.

[0082] With reference to FIG. **16**, a fluid is directed into system **1600** in parallel through the two inlets **1602**. Inlet **1602** may be a tube or a needle. For example, the tube or needle may have a diameter of one millimeter. Instead of feeding the fluid or gas directly into a wide flow chamber with a single continuous space, each of the inlets **1602** feeds the fluid or gas into a separate channel **1608** that directs the fluid or gas to flow directly above a single column of sensor banks **1606**. The

channels **1608** may be formed by stacking together a top plate and a gasket with dividers **1610** that divide the chamber into channels, and then mounting them on top of the chip. Once the fluid or gas flows through the channels **1608**, the fluid or gas is directed up in parallel through the two outlets **1604** and out of system **1600**.

[0083] FIG. **17** illustrates the top view of a nanopore based sequencing system **1700** with another improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface. Similar to system **1600**, the flow chamber in system **1700** includes a horseshoe-shaped flow channel **1708**, but horseshoe-shaped flow channel **1708** directs the fluids to flow directly above two columns (or two rows) of sensor banks **1706**. The width of the channel is about 3.5 millimeters. As shown in FIG. **17**, system **1700** includes an inlet **1702** and an outlet **1704**.

[0084] Both system **1700** and system **1700** allow the fluids to flow more evenly on top of all the sensors on the chip surface. The channel width is configured to be narrow enough such that capillary action has an effect. More particularly, the surface tension (which is caused by cohesion within the fluid) and adhesive forces between the fluid and the enclosing surfaces act to hold the fluid together, thereby preventing the fluid or the air bubbles from breaking up and creating dead zones. Therefore, when the width of a sensor bank is narrow enough, each of the horseshoe-shaped flow channels may flow the fluids directly above two or more columns (or two or more rows) of sensor banks. In this case, system **1700** may be used. When the width of a sensor bank is not narrow enough, then each of horseshoe-shaped flow channels may flow the fluids directly above one column (or one row) of sensor banks only. In this case, system **1600** may be used.

[0085] In some embodiments, the nanopore based sequencing system includes an improved flow chamber having a serpentine fluid flow channel that directs the fluids to traverse over different sensors of the chip along the length of the channel. FIG. **18** illustrates the top view of a nanopore based sequencing system **1800** with an improved flow chamber enclosing a silicon chip that allows liquids and gases to pass over and contact sensors on the chip surface. The flow chamber includes a serpentine or winding flow channel **1808** that directs the fluids to flow directly above a single column (or a single row) of sensor banks **1806** from one end of the chip to the opposite end and then directs the fluids to repeatedly loop back and flow directly above other adjacent columns of sensor banks until all of the sensor banks have been traversed at least once. As shown in FIG. **18**, system **1800** includes an inlet **1802** and an outlet **1804** and the serpentine or winding flow channel **1808** directs a fluid to flow from the inlet **1802** to the outlet **1804** over all 16 sensor banks **1806**.

[0086] With reference to FIG. **18**, a fluid is directed into system **1800** through inlet **1802**. Inlet **1802** may be a tube or a needle. For example, the tube or needle may have a diameter of one millimeter. Instead of feeding the fluid or gas directly into a wide flow chamber with a single continuous space, inlet **1802** feeds the fluid or gas into a serpentine flow channel **1808** that directs the fluid or gas to flow directly above the columns of sensor banks **1806** serially connected together through serpentine flow channel **1808**. The serpentine channel **1808** may be formed by stacking together a top plate and a gasket with dividers **1810** that divide the chamber into the serpentine channel, and then mounting them on top of the chip. Once the fluid or gas flows through the serpentine channel **1808**, the fluid or gas is directed up through outlet **1804** and out of system **1800**.

[0087] System **1800** allows the fluids to flow more evenly on top of all the sensors on the chip surface. The channel width is configured to be narrow enough such that capillary action has an effect. More particularly, the surface tension (which is caused by cohesion within the fluid) and adhesive forces between the fluid and the enclosing surfaces act to hold the fluid together, thereby preventing the fluid or the air bubbles from breaking up and creating dead zones. For example, the channel may have a width of 1 millimeter or less. The narrow channel enables controlled flow of the fluids and minimizes the amount of remnants from a previous flow of fluids or gases.

[0088] FIG. **19A** illustrates an exemplary view of one embodiment of a nanopore based sequencing system **1900** with a serpentine flow channel. FIG. **19B** illustrates the various components that are

laminated together to form nanopore based sequencing system **1900**. System **1900** includes various components, including a printed circuit board **1901**, a nanopore array chip **1902**, a gasket **1904** with dividers **1903**, a backing plate **1907**, a counter electrode **1906** on the underside of backing plate **1907**, a flexible flat circuit **1916** connecting to counter electrode **1906**, an inlet **1908**, an outlet **1910**, a spring plate **1912**, and a plurality of fastening hardware **1914**. The serpentine flow channel is the space formed between backing plate **1907**, gasket **1904**, and nanopore array chip **1902**.

[0089] FIG. **20A** illustrates the top side view of a backing plate and a flexible flat circuit that is connected to the counter electrode (not visible) located on the bottom side of the backing plate. FIG. **20B** illustrates the same unit **2000** as shown in FIG. **20A** when the backing plate is flipped upside down. As shown in this figure, the counter or common electrode **1906** has a serpentine, spiral, or winding shape. Referring back to FIG. **19B**, the counter electrode's serpentine shape matches with the serpentine channel of gasket **1904**, such that the counter electrode is positioned directly above the sensor banks without being blocked by the dividers **1903** of the gasket. The dividers **1903** are disposed between the sensor banks so that the dividers do not block the flow of the fluids or gases over the sensor banks.

[0090] FIG. **20C** illustrates the various components of unit **2000** that are laminated together. Unit **2000** includes a dielectric layer **2002**, a counter electrode **1906** on a film **2004**, a reference electrode **2006**, a reference electrode **2008**, a flexible flat circuit **1916**, and a backing plate **1907**.

[0091] FIG. **19B** and FIG. **20C** illustrate that the flow channel is formed by laminating a backing plate with the counter electrode, a gasket, and the silicon chip together. However, the backing plate with the counter electrode and the gasket may be integrated together as a single unit made of the electrode material, and the unit is machined to form the serpentine flow channel.

[0092] Besides the geometry and dimensions of the flow chamber, other features may also facilitate a more even flow of the fluids on top of all the sensors on the chip surface. FIG. **21A** illustrates a cross sectional view of a flow channel **2100** with sharp edges or sharp corners that may trap fluids more easily. FIG. **21B** illustrates a cross sectional view of a flow channel **2102** that has a curved roof **2103** and a D-shaped cross-sectional geometry. The sharp edges or sharp corners are replaced by round and smooth surfaces. FIG. **21C** illustrates a cross sectional view of another flow channel **2106** that has a curved roof **2107**. FIG. **22** illustrates a side view of a nanopore based sequencing system **2200** with flow channels having a D-shaped cross sectional geometry.

[0093] Another factor that affects the flow of the fluids on top of all the sensors on the chip surface is the height of the flow channel. For example, the height of the flow channel should be limited to one millimeter or below. In one embodiment, the height of the flow channel is 0.25 millimeters. Other factors that affect the flow of the fluids on top of all the sensors on the chip surface include the surface characteristics of the surfaces defining the flow channel, the flow rate of the fluids, the pressure of the fluid and the gases, and the like.

[0094] FIG. **23** is a diagram illustrating an embodiment of a molded flow channel component. In some embodiments, the component shown in FIG. **23** forms at least a portion of a flow channel shown in FIG. **18**. As shown in FIG. **19B**, gasket **1904** with dividers **1903** must be carefully aligned with counter electrode **1906** on the underside of backing plate **1907** during assembly to correctly assemble the flow channel. Additionally, as shown in FIG. **20C**, dielectric layer **2002**, counter electrode **1906**, film **2004**, and backing plate **1907** must be carefully aligned during assembly. Requiring such a large number of components to be aligned together during assembly introduces complexities that may increase manufacturing cost and risk of error. Therefore, there exists a need for an efficient and reliable way to form the flow channel/chamber of a nanopore system.

[0095] Molded flow channel component **2302** includes molded portion **2304** and counter electrode portion **2306**. When molded flow channel component **2302** is placed on a nanopore array chip (e.g., chip **1902**) and secured, the serpentine void shown in molded flow channel component **2302** (e.g., counter electrode portion **2306** is exposed through the serpentine void of the molded portion **2304**) becomes a flow chamber of a fluid flow channel that directs fluids to traverse over different sensors

of the nanopore array chip. Molded portion **2304** includes shown raised dividers that guide the fluid flow along the length of the serpentine channel. Molded flow channel component **2302** includes inlet **2312** and outlet **2314**. Both inlet **2312** and outlet **2314** include a tubular channel that provides fluid/gas flow in or out of the fluid flow chamber/channel. The inlet/outlet tubular channels are formed at least in part by molded portion **2304** and pass through counter electrode portion **2306**.

[0096] Molded portion **2304** has been molded over counter electrode portion **2306**. For example, counter electrode portion **2306** was placed inside a mold and molding material is injected into the mold to form molded portion **2304** over counter electrode portion **2306** to create molded flow channel component **2302**. An example material of molded portion **2304** is an elastomer. An example of counter electrode portion **2306** is a metal (e.g., titanium nitride sputtered/coated stainless steel).

[0097] FIG. **24** is a diagram illustrating an embodiment of a counter electrode insert. In some embodiments, counter electrode insert **2400** becomes counter electrode portion **2306** of FIG. **23** after assembly. Counter electrode insert **2400** may be made of metal (e.g., stainless steel, steel, aluminum, etc.), semiconductor material (e.g., doped silicon) or other conductive material that may be rigid or flexible (e.g., foil). In some embodiments, counter electrode insert **2400** has been coated and/or sputtered with an electrically conductive material. For example, titanium nitride (TiN) has been sputtered on to a base material (e.g., glass, stainless steel, metal, silicon, etc.) to coat counter electrode insert **2400**. This allows the portion of the counter electrode insert **2400** that is exposed inside a flow channel of molded flow channel component **2302** (e.g., portion exposed to chemical reagents) to be titanium nitride. In some embodiments, TiN is a preferred material because of its certain beneficial electrochemical properties, its general availability in and compatibility with existing standard semiconductor manufacturing processes, its compatibility with the biological and chemical reagents, and its relative low cost.

[0098] Counter electrode insert **2400** includes cutouts **2402** and **2404** that allow counter electrode insert **2400** to be aligned and stabilized inside an injection mold. Cutouts **2402** and **2404** are on tab ends of counter electrode insert **2400** that are removed (e.g., snapped off) after molding and discarded from counter electrode insert **2400** to form counter electrode portion **2306** of FIG. **23**. Cutouts **2406** and **2408** allow the tab ends to be more easily removed (e.g., snapped off) in the direction of the length of the cutouts **2406** and **2408**. The tab ends allow easier alignment and stabilization inside a mold as well as easier handling of the component during molding. Cutout **2410** corresponds to inlet **2312** of FIG. **23** and cutout **2412** corresponds to outlet **2314** of FIG. **23** that allows a fluid/gas to pass through counter electrode insert **2400** via tubular channels and in/out of a fluid flow chamber/channel. The other cutouts shown in counter electrode insert **2400** couple counter electrode insert **2400** with a molding material by allowing the molding material to flow through and remain molded in the cutouts to secure the coupling between the counter electrode insert **2400** and the molding material. These other cutouts are placed in locations on the counter electrode insert **2400** to avoid surface portions that are to be exposed inside the fluid chamber/channel. A laser cutter may be utilized as well to produce the shown cutouts. In various embodiments, insert **2400** may have been stamped, chemically etched or cut using a water-jet to produce the shown cutouts.

[0099] FIG. **25** is a diagram illustrating an embodiment of a mold for a molded flow channel component. In some embodiments, mold **2500** is utilized to produce molded flow channel component **2302** of FIG. **23**. A transparent view of mold **2500** is shown to illustrate the internal structure of mold **2500**. Mold **2500** includes a plurality of sectional pieces that are joined together when molding a component. These pieces may be separated after molding to extract the molded component. Counter electrode insert **2400** has been placed inside mold **2500**. Mold **2500** includes tube features that are inserted in cutouts **2410** and **2412** of counter electrode insert **2400** to produce inlet/outlet tubular channels of molded flow channel component **2302**. Counter electrode insert

2400 is secured inside mold **2500** via its tab ends, one or more cutouts and features of mold **2500** contacting a surface of counter electrode insert **2400**. A molding material is injected into mold **2500** to injection mold the molding material around counter electrode insert **2400** and in the shape of mold **2500**. In various embodiments, molding material may be an elastomer, rubber, silicone, polymer, thermoplastics, plastics, or any other injection moldable material. One or more cutouts/holes on counter electrode insert **2400** may be filled with the molding material to couple the molding material with counter electrode insert **2400**. In some embodiments, the molding material injected into mold **2500** becomes molded portion **2304** of FIG. **23**. In various other embodiments, other molding techniques such as liquid injection molding (LIM), transfer molding, or compression molding are utilized to produce molded flow channel component **2302** of FIG. **23**. [0100] FIG. **26** is a diagram illustrating an embodiment of a molded flow channel component removed from a mold. Component **2600** has been removed from mold **2500** after injection molding. After removing (e.g., bending/snapping off) shown end tabs of counter electrode insert **2400**, component **2600** becomes molded flow channel component **2302** of FIG. **23**. The four by three array of openings/holes on top of molded portion **2304** exposes surfaces of counter electrode insert **2400**. In some embodiments, this array of openings/holes is an artifact resulting from the twelve features/pins of mold **2500** that are used to hold and stabilize counter electrode insert **2400** in place in the mold during molding. In some embodiments, at least some of these openings/holes of the array are utilized to make electrical contact with a counter electrode portion. For example, a spring contact is placed in some of the openings/holes of the array and connected to an electrical potential source of a circuit board (e.g., shown in FIG. **27**).

[0101] FIG. **27** is a diagram illustrating a portion of an embodiment of a nanopore based sequencing system utilizing a molded flow channel component. Molded flow channel component **2302** has been placed on top of a nanopore array chip that is mounted on circuit board **2702**. Spring contact wire component **2704** is to be placed in the array of openings/holes on top of molded flow channel component **2302** to make electrical and physical contact with its counter electrode portion. The other end of spring contact wire component **2704** is to be connected to circuit board **2702** to provide an electrical potential source for its counter electrode. In some embodiments, the molded flow channel component **2302** and electrical contact of spring contact wire component **2704** of the counter electrode are secured and clamped via a clamp plate (e.g., without use of adhesives) that clamps spring contact wire component **2704** to molded flow channel component **2302**. In some embodiments, the clamp plate also clamps molded flow channel component **2302** to the nanopore array chip to provide the compression required to seal and couple them together in creating a flow channel/chamber between molded flow channel component **2302** and the nanopore array chip. Encapsulate berm **2706** encapsulates wire bonds between electrical terminals of the nanopore array chip and circuit board **2702**.

[0102] FIG. **28** is a diagram illustrating an embodiment of a clamping of nanopore based sequencing system components. The views shown in FIG. **28** have been simplified to illustrate the embodiment clearly. View **2800** shows an overview of the components of FIG. **27** that have been clamped together.

[0103] Clamping plate **2802** clamps together molded flow channel component **2302** over a nanopore array chip. Clamping plate **2802** may be fastened and clamped to circuit board **2702** via screws or other coupling mechanisms. Clamping plate **2802** includes holes to accommodate and allow inlet **2312** and outlet **2314** to pass through clamping plate **2802**. Clamping plate **2802** clamps the molded flow channel component to the nanopore array chip to provide the compression required to seal and couple them together in creating a flow channel/chamber without the use of adhesives and/or permanent/physical bonding to couple together the molded flow channel component to the nanopore array chip. Spring contact wire component **2704** is not shown to simplify the diagram but exists in the embodiment under clamping plate **2802**. Clamping plate **2802** also secures the spring contact wire component to the counter electrode of the molded flow

channel component via compression.

[0104] View **2801** shows a cutaway view of the embodiment shown in view **2800**. Clamping plate **2802** is secured to circuit board **2702** and clamps together nanopore array chip **2804** with counter electrode portion **2306** and molded portion **2304** of molded flow channel component **2302**. The gap between counter electrode portion **2306** and nanopore array chip **2804** is a part of a flow chamber/channel that holds and directs the fluids to traverse over different sensors of nanopore array chip **2804**. The clamping force/pressure seals contact between molded portion **2304** and nanopore array chip **2804**. In some embodiments, the material of molded portion **2304** is compliant to provide the seal under clamping force/pressure.

[0105] FIG. **29** is a diagram illustrating an embodiment of encapsulated wire bonds. Wire bonds **2906** electrically connect nanopore based sequencing chip **2904** to circuit board **2902**. Wire bonds **2906** are protected by encapsulant **2908**. Although only two wire bonds have been shown to simplify the diagram, other wire bonds **2906** all around the perimeter of chip **2904** may be utilized to electrically connect chip **2904** to circuit board **2902**. In order to protect wire bonds **2906** from damage and wetting, encapsulant **2908** (e.g., adhesive) is utilized to encapsulate wire bonds **2906** by applying a ring of encapsulant around the perimeter of chip **2904**. In some embodiments, encapsulant **2908** is the encapsulant shown as surrounding a nanopore chip in other figures. However, the applied encapsulant cannot encroach on portions of the chip where one or more flow channel components are to be coupled to the chip.

[0106] Thus care must be taken to minimize the encroachment of the encapsulant onto the surface of the chip in order to maximize the area available on the chip surface for coupling the one or more flow channel components onto the chip. In various embodiments, the one or more flow channel components may be coupled to the chip surface by room temperature bonding, adhesive bonding, and/or compression bonding. During manufacturing, two distinct steps may need to be performed—first, encapsulating the wire bonds with careful attention paid to preventing undesired encroachment of the encapsulant onto the chip and then coupling and sealing the one or more flow channel components onto the chip.

[0107] FIG. **30** is a diagram illustrating an embodiment of encapsulating wire bonds together with one or more flow channel components in a single manufacturing step. For example, rather than encapsulating wire bonds between a nanopore chip and a circuit board in a separate step from securing one or more flow channel components to the nanopore chip, a single application of encapsulant **3004** is utilized to both encapsulate the wire bonds and couple/secure the one or more flow channel components to the chip. In some embodiments, without first encapsulating wire bonds, flow channel component **3002** is placed on the nanopore chip. Then, a single application of encapsulant (e.g., adhesive) is applied to simultaneously encapsulate the wire bonds and seal the perimeter of flow channel component **3002** to the nanopore based sequencing chip. By combining two manufacturing steps into one encapsulation step, manufacturing becomes more efficient and simplified. Additionally, the chip encroachment restriction on the application of the encapsulant is eliminated because the flow channel component is already placed on the chip and limits the flow of the encapsulant on undesired portions of the chip. Examples of flow channel component **3002** include various components placed on a nanopore cell to form a chamber/flow channel over the sensors of the chip. For example, various gaskets, plates, and reference electrodes described in various embodiments herein may be included in flow channel component **3002**. In some embodiments, flow channel component **3002** includes molded flow channel component **2302** of FIG. **23**.

[0108] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There are many alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

[0109] All publications, patents, patent applications, and/or other documents cited in this

application are incorporated by reference in their entirety for all purposes to the same extent as if each individual publication, patent, patent application, and/or other document were individually indicated to be incorporated by reference for all purposes.

Claims

1. A nanopore based sequencing system, the system comprising: a nanopore sensor array comprising a plurality of nanopore sensors, each nanopore sensor comprising a cell and a working electrode; a counter electrode; a flow channel disposed over the nanopore sensor array; an inlet configured to deliver the fluid into the flow channel, wherein the inlet is located proximate a first side of the nanopore sensor array; and an outlet configured to deliver fluid out of the flow channel, wherein the outlet is also located proximate the first side of the nanopore sensor array.
2. The nanopore based sequencing system of claim 1, wherein the nanopore sensory array comprises one million nanopore sensors.
3. The nanopore based sequencing system of claim 1, wherein the cell is configured to support a membrane and a nanopore disposed in the membrane.
4. The nanopore based sequencing system of claim 3, wherein the nanopore is a protein nanopore.
5. The nanopore based sequencing system of claim 4, wherein the protein nanopore is attached to a polymerase.
6. The nanopore based sequencing system of claim 5, wherein the polymerase is associated with a nucleic acid molecule to be sequenced.
7. The nanopore based sequencing system of claim 6, wherein the nucleic acid molecule is circular.
8. The nanopore based sequencing system of claim 1, wherein the flow channel is horseshoe shaped.
9. The nanopore based sequencing system of claim 1, wherein the flow channel is serpentine shaped.
10. The nanopore based sequencing system of claim 1, wherein the flow channel comprises at least one fluid guide.
11. A nanopore based sequencing system, the system comprising: a nanopore sensor array comprising a plurality of nanopore sensors, each nanopore sensor comprising a cell and a working electrode, wherein the cell is configured to support a membrane and a nanopore disposed in the membrane, wherein the nanopore is attached to a polymerase, wherein the polymerase is associated with a nucleic acid molecule to be sequenced, wherein the nucleic acid molecule is circular; a counter electrode; a flow channel disposed over the nanopore sensor array; an inlet configured to deliver the fluid into the flow channel, wherein the inlet is located proximate a first side of the nanopore sensor array; and an outlet configured to deliver fluid out of the flow channel, wherein the outlet is also located proximate the first side of the nanopore sensor array.
12. The nanopore based sequencing system of claim 11, wherein the nanopore sensory array comprises one million nanopore sensors.
13. The nanopore based sequencing system of claim 11, wherein the flow channel is horseshoe shaped.
14. The nanopore based sequencing system of claim 11, wherein the flow channel is serpentine shaped.
15. The nanopore based sequencing system of claim 11, wherein the flow channel comprises at least one fluid guide.
16. A nanopore based sequencing system, the system comprising: a nanopore sensor array comprising a plurality of nanopore sensors, each nanopore sensor comprising a cell and a working electrode, wherein the cell is configured to support a membrane and a nanopore disposed in the membrane, wherein the nanopore is attached to a polymerase, wherein the polymerase is associated with a nucleic acid molecule to be sequenced, wherein the nucleic acid molecule is circular; a

counter electrode; a flow channel disposed over the nanopore sensor array, wherein the flow channel is horseshoe shaped; an inlet configured to deliver the fluid into the flow channel, wherein the inlet is located proximate a first side of the nanopore sensor array; and an outlet configured to deliver fluid out of the flow channel, wherein the outlet is also located proximate the first side of the nanopore sensor array.

17. The nanopore based sequencing system of claim 16, wherein the nanopore sensory array comprises one million nanopore sensors.

18. A nanopore based sequencing system, the system comprising: a nanopore sensor array comprising a plurality of nanopore sensors, each nanopore sensor comprising a cell and a working electrode, wherein the cell is configured to support a membrane and a nanopore disposed in the membrane, wherein the nanopore is attached to a polymerase, wherein the polymerase is associated with a nucleic acid molecule to be sequenced, wherein the nucleic acid molecule is circular; a counter electrode; a flow channel disposed over the nanopore sensor array, wherein the flow channel is serpentine shaped; an inlet configured to deliver the fluid into the flow channel, wherein the inlet is located proximate a first side of the nanopore sensor array; and an outlet configured to deliver fluid out of the flow channel, wherein the outlet is also located proximate the first side of the nanopore sensor array.

19. The nanopore based sequencing system of claim 18, wherein the nanopore sensory array comprises one million nanopore sensors.
